

JAYDEN CHAN

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Education

University of Waterloo

Waterloo, Ontario

Bachelor of Computer Science

Wilfrid Laurier University

Waterloo, Ontario

Bachelor of Business Administration

Technical Skills

Languages: Python, C/C++, Java, Javascript, Racket, HTML/CSS, SQL

Frameworks & Libraries: TensorFlow, OpenCV, MediaPipe, Flask, NumPy, Pandas, Librosa, Matplotlib

Tools & Platforms: Git, GitHub, Linux/Bash, FFmpeg, SceneKit, Metal, RESTful APIs

Concepts: Computer Vision, Digital Signal Processing, Technical Analysis, Agile Development

Experience

Software Development Intern

July 2024 – October 2024

FuturIQ Inc.

Brampton, Ontario

- **Increased user engagement** with AI interfaces by developing an **interactive 3D digital human** in **Unity** using **MetaHuman Creator** to simulate realistic, human-like interactions.
- **Streamlined social media workflows** by engineering a **customized chatbot** in **Python** designed to act as an automated advisor for content optimization.
- **Enhanced web accessibility** for AI tools by building an **interactive HTML/CSS site** featuring a **beginner-friendly GUI** and seamless **chatbot integration**.
- **Scaled organizational AI literacy** by leading workshops on **GPT tools** for diverse audiences, demonstrating **practical AI applications** and implementation strategies.

Projects

Stock Analysis & Portfolio Optimization Tool

February 2026

- **Generated real-time buy/sell signals** for diverse equities by developing a **Flask-based web platform** that analyzes technical indicators across live market data.
- **Quantified market volatility** within 1-year historical datasets by implementing **mathematical algorithms** including **RSI** and **Bollinger Bands** using **Pandas** and **NumPy**.
- **Optimized asset allocation** through a **custom algorithmic scoring system** by engineering a feature that dynamically distributes capital based on user-defined risk parameters.
- **Reduced data latency** for market predictions by designing and deploying **RESTful API endpoints** to handle high-frequency requests for trend scanning.

Music Sheet Converter

February 2026

- **Automated the music transcription process** from multi-format audio/video sources by engineering an **AI-driven system** that outputs high-fidelity **sheet music** and **instrument tabs**.
- **Improved translation accuracy** of complex tutorials into **playable notation** by implementing a **multimodal feature** that synchronizes visual cues with audio instruction.
- **Standardized audio output formats** including **MIDI** and **PDF** by implementing **advanced audio processing** to identify precise pitch and rhythmic intervals.

AI Facial Geometry Analysis

January 2026 – Present

- **Achieved real-time 3D geometry analysis** of facial structures by building a **computer vision application** in **Python** utilizing **MediaPipe** to track **478 facial landmarks**.
- **Quantified structural facial aesthetics** by developing a **mathematical engine** using **NumPy** to calculate metrics such as **facial symmetry**, **canthal tilt**, and **golden ratio alignment**.
- **Automated objective harmony measurements** by implementing **vector calculus algorithms** to analyze spatial proportions and facial ratios.

Leadership / Extracurricular

St. Edmund Campion

September 2024 - June 2025

Artificial Intelligence Club President

Brampton, Ontario

- Mentored and taught over 20 students from grade 9-11 the basics of artificial intelligence.
- Developed projects using prediction models (tensorflow), computer vision models (OpenCV) and customizable chatbots.